

Question	Answer	Marks
4(a)(iii)	any three from: <i>eukaryote because presence of</i> (true) nucleus / nuclear envelope / nuclear membranes ; mitochondrion ; <i>if mp1 and 2 not gained, allow double membrane-bound structures</i> nucleolus ; chromatin ; (X) has 9+2 pattern of microtubules ; A ref. to cytoskeleton / microtubules X / AW, covered by cell (surface) membrane ; <u>endoplasmic reticulum</u> ; I rough / smooth I ER / SER / RER A Golgi body <i>not prokaryote because</i> absence of cell wall ; I does not have peptidoglycan cell wall (size) greater than 5 µm ; A larger than (typical) prokaryote / bacterial cell A valid calculated size using scale bar <i>if stated as a prokaryote, read whole response and if all ideas refer correctly to eukaryotic structures, allow max 1</i>	3
4(b)(i)	pathogen ; I parasite	1
4(b)(ii)	<i>falciparum / malariae / ovale / vivax ; A knowlesi correct spelling</i>	1

Question	Answer	Marks
4(c)	any three from: <i>similarities</i> <u>vector</u>, qualified ; e.g. <u>vector</u> is, an insect / a fly both, are <u>vector</u>-borne / are carried by a <u>vector</u> both, need / involve, a <u>vector</u> (vector / carrier / insect) feeds / AW, on (human) blood ; A both diseases transmitted through blood <i>differences</i> <i>malaria</i> (vector) is, a mosquito / <i>Anopheles</i> ; ora (tsetse fly) not a mosquito R mosquito / <i>Anopheles</i>, versus insect (only) female (<i>mosquitos</i>), feed on blood / transmit disease / are vectors ; <i>(information given for sleeping sickness is that male and female tsetse flies feed on blood)</i>	3